

C Programming

BCA — Semester 2 — 2021 — Answer Sheet

Subject Code: CACS151

Group A

Attempt all the questions. [10×1 = 10] — Note: Groups B and C were not found in the source PDF and are omitted from this answer sheet.

1. Among unary operation which operator represents increment?

[1]

Answer: Correct Answer: b) ++

The **increment operator (++)** is a unary operator in C that increases the value of a variable by 1. It can be used in prefix form (++x) or postfix form (x++). The decrement operator (--) decreases a value by 1. Options -+, !+, and - - (decrement) are not correct for increment.

2. If the function returns no value, then it is called.....

[1]

Answer: Correct Answer: d) Void function

A **void function** in C is a function declared with the void return type, meaning it does not return any value to the calling function. It performs operations but does not send back a result. Functions that return data are declared with specific data types like int, float, or char.

3. What are the C functions used to read or write a file in Binary mode?

[1]

Answer: Correct Answer: b) fread(), fwrite()

The fread() and fwrite() functions are used for binary file I/O in C. fread() reads a specified number of bytes from a binary file into memory, while fwrite() writes a block of data to a binary file. fprintf() and fscanf() are for text mode, and getw()/putw() are for reading/writing integers.

4. An entire array is always passed byto a called function.

[1]

Answer: Correct Answer: b) call by reference

In C, when an array is passed to a function, what is actually passed is the **address of the first element** (a pointer). This means the array is passed by reference, not by value. Any modifications made to the array elements inside the function affect the original array in the calling function.

5. What is actually passed if you pass a structure variable to a function?

[1]

Answer: Correct Answer: c) Starting address of structure variable

When a structure variable is passed to a function, the entire structure is typically passed by value (a copy is made). However, if a pointer to the structure is passed, then the **starting address** of the structure is passed. Given the context of standard C behavior with pointers, the starting address represents the reference-passing mechanism used for efficiency with large structures.

6. Choose a correct statement about C Escape Sequences.

[1]

Answer: Correct Answer: d) All the above

All listed escape sequences are correct in C:

- `'` outputs a single quote
- `"` outputs a double quote
- `\\` produces one visible backslash

These are standard escape sequences defined in the C language for formatting output and handling special characters within strings and character constants.

7. What is the keyword used to define a c macro?

[1]

Answer: Correct Answer: c) define

The `#define` preprocessor directive is used to define macros in C. Macros are symbolic names for constants or code snippets that are replaced by the preprocessor before compilation. Example: `#define PI 3.14159` or `#define SQUARE(x) ((x)*(x))`. Unlike functions, macros are expanded inline.

8. What is the first step in c program building process.?

[1]

Answer: Correct Answer: d) Preprocessing

The C program building process follows this order:

1. **Preprocessing:** Handles directives like `#include`, `#define`, conditional compilation
2. **Compiling:** Translates preprocessed C code into assembly
3. **Assembling:** Converts assembly to object code (machine code)
4. **Linking:** Combines object files and libraries into executable

Therefore, preprocessing is always the first step.

9. Choose a correct C statement about String functions?

[1]

Answer: Correct Answer: b) strcmp("abc", "bcd") returns a negative number

strcmp() compares strings lexicographically using ASCII values. Since 'a' (97) < 'b' (98), "abc" is less than "bcd", so it returns a negative number. Option (a) is incorrect because strrev() modifies in place and returns the reversed string pointer — "abcD" reversed would be "Dcba". Option (c) is wrong because "234" > "123" so strcmp would return positive.

10. Choose a syntax for C Ternary Operator from the list.

[1]

Answer: Correct Answer: a) condition? Expression1 : expression2

The **ternary operator (? :)** is C's conditional operator. It evaluates the condition; if true, returns Expression1, otherwise returns Expression2. Example: max = (a > b) ? a : b; assigns the larger of a and b to max. It is a shorthand for simple if-else statements and is right-associative.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.